

Soil amendments: compost and biochar

Spatial scale: vineyard

About this sustainable management strategy

The application of organic soil amendments, such as compost and biochar, is a widely used practice to improve soil health and support long-term vineyard resilience. Unlike synthetic mineral fertilisers, these amendments contribute organic matter to the soil, enhancing nutrient cycling, biological activity, and soil structure. They can also improve water retention, increase infiltration, and support the accumulation of soil organic carbon.

Composts, including green waste compost, vermicompost, and composted manure, provide nutrients and organic matter that support soil fertility and microbial activity. Biochar is a stable, carbon-rich material produced through the pyrolysis of organic biomass under low-oxygen conditions. When incorporated into soil, these amendments can improve soil structure, increase water-holding capacity, and contribute to long-term carbon storage.

Key ecosystem services

- Soil health and fertility (*improved soil structure, nutrient cycling and organic matter content*)
- Erosion control (*slope stabilisation*)
- Water retention (*reduced surface runoff*)
- Water quality
- Carbon sequestration
- Biodiversity enhancement (*above-and below-ground*)
- Natural pest and pathogen control (*support for beneficial insects and natural enemies*)
- Grape production (*improved vine growth and yield*)

Considerations for successful implementation

Amendment selection and application rates

The effects of biochar and compost depend on their composition, quality, and application rate. Selecting appropriate materials and adapting application rates to soil conditions and vineyard objectives are important for achieving the desired improvements in soil fertility and water management.

Nutrient management and vine response

Organic amendments influence nutrient availability and soil biological activity. Their use should be integrated into broader nutrient management strategies to maintain balanced vine growth and avoid excessive vegetative development or nutrient imbalances.

Monitoring of soil and vineyard performance

The response of soils and vines to organic amendments varies according to soil type, climate, and management practices. Periodic assessment of soil properties, vine performance, and nutrient status provides a basis for adapting management practices and optimising long-term benefits.